



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 10: Cost and Revenue Projection

BSS 310 Engineering Management 2026

Lecturer: Prof. Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- Constructing network diagrams from a WBS
- Calculation of the early start and finish dates using the forward pass
- Calculation of the latest start and latest finish dates using the backward pass
- Identification of the critical path in a network diagram and
- Calculation of the total float and free float of activities

Learning Outcomes for the Current Lecture

The expected learning outcomes include:

- Understanding the primary reasons for cost and revenue projection.
- Understanding some significant cost components commonly used in cost and revenue projection.
- Ability to apply the “bottom-up” or “start to completion” approach of operational activities, in cost and revenue projection.
- Being able to analyse and evaluate cost and revenue projections into the future.
- Ability to create, analyse, and evaluate best, most probable optimal, and worst-case cost and revenue scenarios.
- Adaptation or application of the entire knowledge to the semester project.

Presentation Outline

- Why Learn about Cost and Revenue Projection?
- Significant Cost Components for Cost and Revenue Projection
- Cost Estimation Techniques
- Cost and Revenue Projection
- Semester Project discussions for best, optimal, and worst case scenarios

Why learn about cost and revenue projection?

What is the primary driving factor for seeking knowledge in cost and revenue projection?

Significant Cost Components for Cost and Revenue Projection

Setup Costs

These are made up of any initial once off cost usually categorised as a fixed cost. Examples include:

- Re-tooling costs
- Installation & Commissioning costs
- Chemical preparation for service rendering
- Initial orientation and training of personnel in service and product systems etc.

Operating Costs

These are costs that bother around the actual act of production or service rendering. They cannot be avoided hence often referred to as direct costs e.g material, machinery, power, labour costs etc.

Variable Costs

Variable costs are costs that change with production volume. They are often referred to as direct costs.

- Material cost
- Labour cost and
- Direct expenses on add-ons e.g. packaging, labelling etc.
- Power consumption

Overhead Costs

These are usually referred to as fixed costs. They mostly form the indirect costs that do not change with production volume e.g. indirect material costs, indirect labour costs and other indirect expenses. This can span across:

- **Factory overhead costs e.g. salaries, maintenance costs etc**
- **Administration overhead costs e.g. costs associated with administering the business; dispatch, legal, insurance and documentation costs.**
- **Marketing overhead costs e.g. cost of adverts and publicity**
- **Distribution overhead costs e.g. networking and logistics cost**

Overhead Costs...Contd.

- Rent
- Utilities
- Insurance
- Office supplies
- Travel
- Accounting, Salaries and wages etc

Overhead Rate

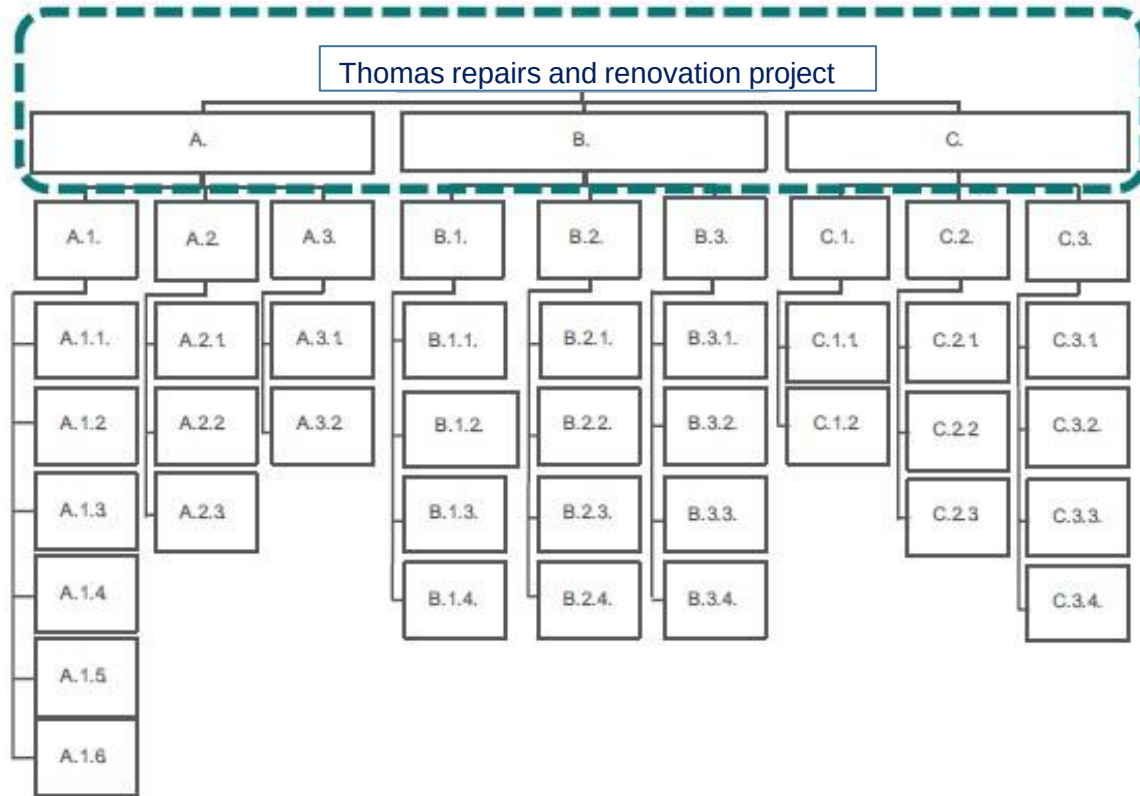
A measure used to monitor and cut-down on frivolous costs:

$$\frac{\text{Indirect costs}}{\text{Direct costs}} \geq 1$$

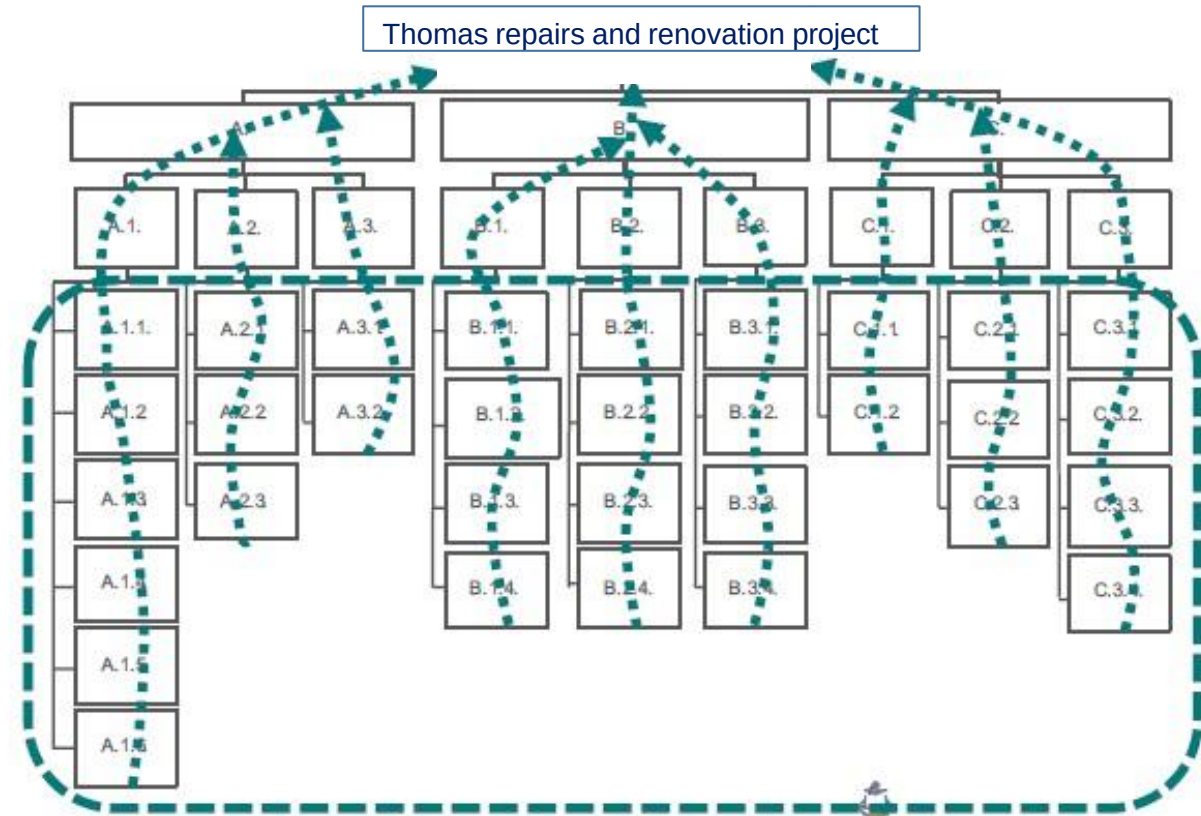
$$\frac{\text{Indirect costs}}{\text{Direct costs}} < 1$$

Cost Estimation Techniques

TOP-DOWN



BOTTOM-UP



Cost Estimation Techniques...Contd..

Top Down Estimation Methods

- Expert judgement
- Analogous estimation
- Parametric estimation
- Three-point estimation

Cost Estimation Techniques...Contd..

- Bottom up Estimation Method

Part B-Section 4: Project Set-up Budget

Cost/Unit/Month

Given the following production/service related information:

Variable Costs

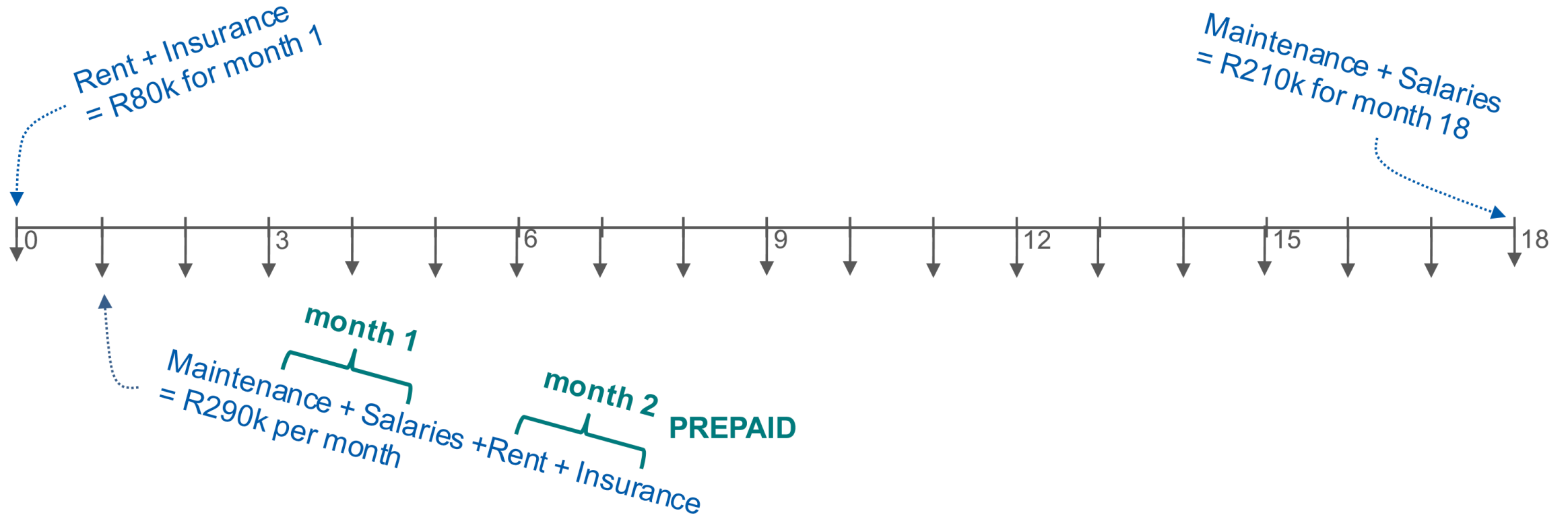
- Materials R300/unit
- Labour R55/unit
- Electricity R25/unit

Fixed Costs

- **Rent @ R60 000/month** *for the production arena, machines and two office space.*
- **Maintenance @ R10 000/month**
- **Insurance @ R20 000/month** *for personnel, machineries and other assets.*
- **Salaries R200 000/month** *for 12 staff members.*

Part B-Section 5: Project Budget for a given period of time

Fixed Costs



Note:

Insurance=R20000; Rent=R60000; Maintenance=R10000 and Salaries=R200000

Month	Fixed cost/month (R)				Variable cost/unit (R)		
Unit cost projection (Production Department)							
	Rent	Maintenance	Insurance	Salaries	Materials	Labour	Electricity
0	60000	-	20000	-			
1	60000	10000	20000	200000	300	55	25
2	60000	10000	20000	200000	300	55	25
3	60000	10000	20000	200000	300	55	25
4	60000	10000	20000	200000	300	55	25
5	60000	10000	20000	200000	300	55	25
6	60000	10000	20000	200000	300	55	27,5
7	60000	10000	20000	200000	300	55	27,5
8	60000	10000	20000	200000	300	55	27,5
9	60000	10000	20000	200000	300	55	27,5
10	60000	10000	20000	200000	300	55	27,5
11	60000	10000	20000	200000	300	55	27,5
12	60000	10000	20000	200000	300	55	27,5
13	60000	10000	20000	200000	300	58,3	27,5
14	60000	10000	20000	200000	300	58,3	27,5
15	60000	10000	20000	200000	300	58,3	29,5
16	60000	10000	20000	200000	300	58,3	29,5
17	60000	10000	20000	200000	300	58,3	29,5
18	-	10000	-	200000	300	58,3	29,5
Total	1080000	180000	360000	3600000	5400	1009,8	490,5
Sum Total	R5220000				R6900,3		
Grand Total	R5226900,30						

Cost projection for 18 mths

Part A-Section 6: Overall Project Cash Flow Analysis (Based on PART B)

- Present consolidated cash flow for the first three years of operation
- Define and quantify worst case, best case and most probable cash flow projections

Cost & Revenue Projection

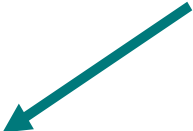
Assumption: Selling price @ R450.00 per unit and increasing at an approximate % rate per month.

Month	Estimated demand ('000)	Planned production volume ('000)	Sales volume ('000)
1	5	5	5
2	6	5	5
3	8	10	8
4	10	10	10
5	13	15	13
6	23	15	19
7	37	15	15
8	55	50	50
9	80	50	50
10	115	50	50
11	120	100	100
12	122	110	110
13	123	120	120
14	124	120	120
15	125	120	120
16	125	120	120
17	125	120	120
18	125	120	120

6% wage increase
per year



erratic electricity
price increases



Month	Selling price	Sales volume ('000)	Monthly revenue	Quarterly revenue	Labour (per unit)	Materials (per unit)	Electricity (per unit)	Monthly COGS	Quarterly COGS	Monthly gross profit	Quarterly gross profit
1	R 450,00	5	R 2 250 000,00		R 55,00	R 300,00	R 25,00	R 1 900 000,00		R 350 000,00	
2	R 450,90	5	R 2 254 500,00		R 55,00	R 300,00	R 25,00	R 1 900 000,00		R 354 500,00	
3	R 451,80	8	R 3 614 414,40	R 8 118 914,40	R 55,00	R 300,00	R 25,00	R 3 040 000,00	R 6 840 000,00	R 574 414,40	R 1 278 914,40
4	R 452,71	10	R 4 527 054,04		R 55,00	R 300,00	R 25,00	R 3 800 000,00		R 727 054,04	
5	R 453,61	13	R 5 896 940,59		R 55,00	R 300,00	R 25,00	R 4 940 000,00		R 956 940,59	
6	R 454,52	19	R 8 635 842,68	R 19 059 837,31	R 55,00	R 300,00	R 27,50	R 7 267 500,00	R 16 007 500,00	R 1 368 342,68	R 3 052 337,31
7	R 455,43	15	R 6 831 406,08		R 55,00	R 300,00	R 27,50	R 5 737 500,00		R 1 093 906,08	
8	R 456,34	50	R 22 816 896,31		R 55,00	R 300,00	R 27,50	R 19 125 000,00		R 3 691 896,31	
9	R 457,25	50	R 22 862 530,11	R 52 510 832,50	R 55,00	R 300,00	R 27,50	R 19 125 000,00	R 43 987 500,00	R 3 737 530,11	R 8 523 332,50
10	R 458,17	50	R 22 908 255,17		R 55,00	R 300,00	R 27,50	R 19 125 000,00		R 3 783 255,17	
11	R 459,08	100	R 45 908 143,35		R 55,00	R 300,00	R 27,50	R 38 250 000,00		R 7 658 143,35	
12	R 460,00	110	R 50 599 955,60	R 119 416 354,12	R 55,00	R 300,00	R 27,50	R 42 075 000,00	R 99 450 000,00	R 8 524 955,60	R 19 966 354,12
13	R 460,92	120	R 55 310 351,47		R 58,30	R 300,00	R 27,50	R 46 296 000,00		R 9 014 351,47	
14	R 461,84	120	R 55 420 972,17		R 58,30	R 300,00	R 27,50	R 46 296 000,00		R 9 124 972,17	
15	R 462,77	120	R 55 531 814,12	R 166 263 137,76	R 58,30	R 300,00	R 29,70	R 46 560 000,00	R 139 152 000,00	R 8 971 814,12	R 27 111 137,76
16	R 463,69	120	R 55 642 877,74		R 58,30	R 300,00	R 29,70	R 46 560 000,00		R 9 082 877,74	
17	R 464,62	120	R 55 754 163,50		R 58,30	R 300,00	R 29,70	R 46 560 000,00		R 9 194 163,50	
18	R 465,55	120	R 55 865 671,83	R 167 262 713,07	R 58,30	R 300,00	R 29,70	R 46 560 000,00	R 139 680 000,00	R 9 305 671,83	R 27 582 713,07

COGS=Cost of Goods Sold

Month	Selling price	Sales volume ('000)	Monthly revenue	Quarterly revenue	Labour (per unit)	Materials (per unit)	Electricity (per unit)	Monthly COGS	Quarterly COGS	Monthly gross profit	Quarterly gross profit
1	R 450,00	5	R 2 250 000,00		R 55,00	R 300,00	R 25,00	R 1 900 000,00		R 350 000,00	
2	R 450,90	5	R 2 254 500,00		R 55,00	R 300,00	R 25,00	R 1 900 000,00		R 354 500,00	
3	R 451,80	8	R 3 614 414,40	R 8 118 914,40	R 55,00	R 300,00	R 25,00	R 3 040 000,00	R 6 840 000,00	R 574 414,40	R 1 278 914,40
4	R 452,71	10	R 4 527 054,04		R 55,00	R 300,00	R 25,00	R 3 800 000,00		R 727 054,04	
5	R 453,61	13	R 5 896 940,59		R 55,00	R 300,00	R 25,00	R 4 940 000,00		R 956 940,59	
6	R 454,52	19	R 8 635 842,68	R 19 059 837,31	R 55,00	R 300,00	R 27,50	R 7 267 500,00	R 16 007 500,00	R 1 368 342,68	R 3 052 337,31
7	R 455,43	15	R 6 831 406,08		R 55,00	R 300,00	R 27,50	R 5 737 500,00		R 1 093 906,08	
8	R 456,34	50	R 22 816 896,31		R 55,00	R 300,00	R 27,50	R 19 125 000,00		R 3 691 896,31	
9	R 457,25	50	R 22 862 530,11	R 52 510 832,50	R 55,00	R 300,00	R 27,50	R 19 125 000,00	R 43 987 500,00	R 3 737 530,11	R 8 523 332,50
10	R 458,17	50	R 22 908 255,17		R 55,00	R 300,00	R 27,50	R 19 125 000,00		R 3 783 255,17	
11	R 459,08	100	R 45 908 143,35		R 55,00	R 300,00	R 27,50	R 38 250 000,00		R 7 658 143,35	
12	R 460,00	110	R 50 599 955,60	R 119 416 354,12	R 55,00	R 300,00	R 27,50	R 42 075 000,00	R 99 450 000,00	R 8 524 955,60	R 19 966 354,12
13	R 460,92	120	R 55 310 351,47		R 58,30	R 300,00	R 27,50	R 46 296 000,00		R 9 014 351,47	
14	R 461,84	120	R 55 420 972,17		R 58,30	R 300,00	R 27,50	R 46 296 000,00		R 9 124 972,17	
15	R 462,77	120	R 55 531 814,12	R 166 263 137,76	R 58,30	R 300,00	R 29,70	R 46 560 000,00	R 139 152 000,00	R 8 971 814,12	R 27 111 137,76
16	R 463,69	120	R 55 642 877,74		R 58,30	R 300,00	R 29,70	R 46 560 000,00		R 9 082 877,74	
17	R 464,62	120	R 55 754 163,50		R 58,30	R 300,00	R 29,70	R 46 560 000,00		R 9 194 163,50	
18	R 465,55	120	R 55 865 671,83	R 167 262 713,07	R 58,30	R 300,00	R 29,70	R 46 560 000,00	R 139 680 000,00	R 9 305 671,83	R 27 582 713,07

Cost and Revenue Projection

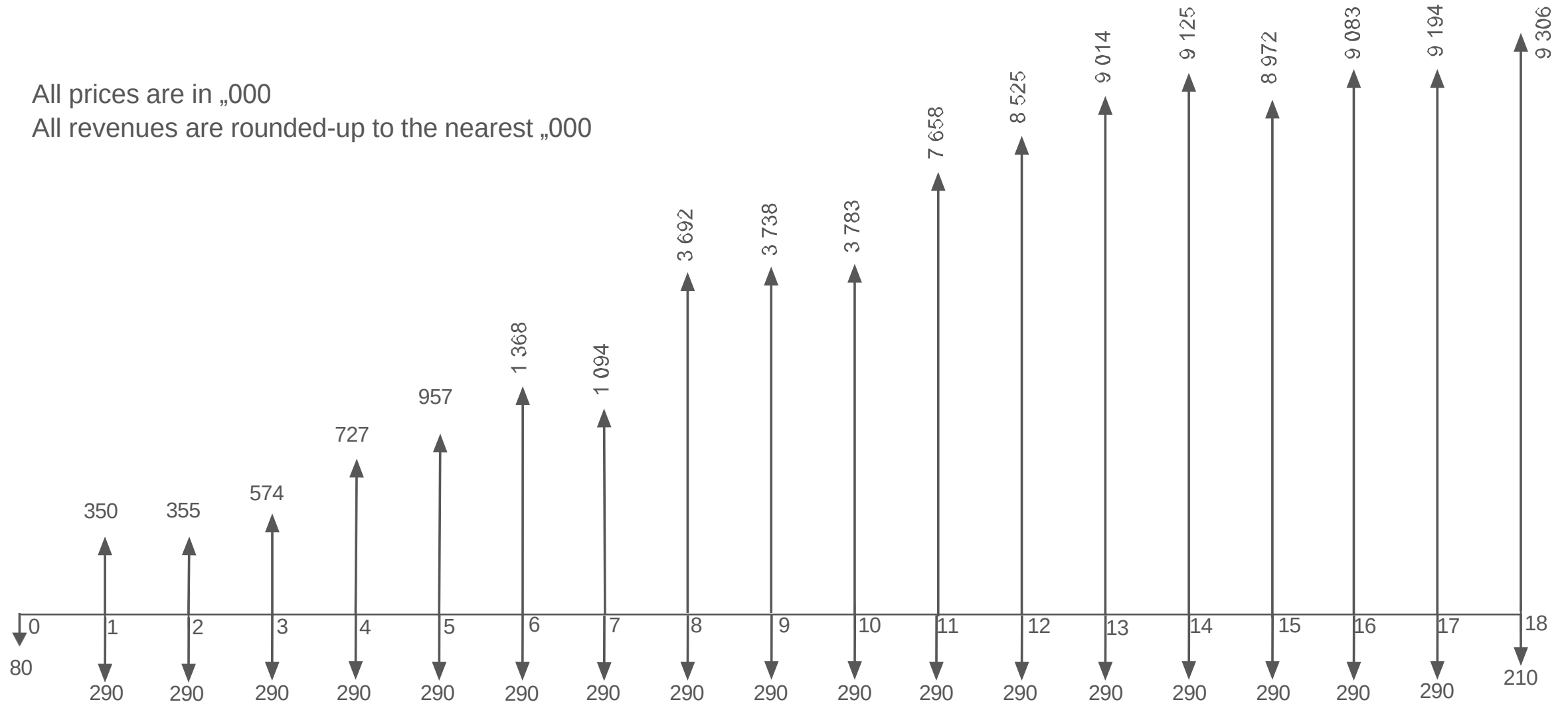
Based on the previous slide of monthly fixed costs given as:
Insurance=R20000; Rent=R60000; Maintenance=R10000 and
Salaries=R200000

	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Quarter 5	Quarter 6
Revenue	R 8 118 914,40	R 19 059 837,31	R 52 510 832,50	R 119 416 354,12	R 166 263 137,76	R 167 262 713,07
Cost of goods sold	R 6 840 000,00	R 16 007 500,00	R 43 987 500,00	R 99 450 000,00	R 139 152 000,00	R 139 680 000,00
Gross profit	R 1 278 914,40	R 3 052 337,31	R 8 523 332,50	R 19 966 354,12	R 27 111 137,76	R 27 582 713,07
Operating expenses	R 950 000,00	R 870 000,00	R 870 000,00	R 870 000,00	R 870 000,00	R 790 000,00
Rent	R 240 000,00	R 180 000,00	R 180 000,00	R 180 000,00	R 180 000,00	R 120 000,00
Maintenance	R 30 000,00	R 30 000,00	R 30 000,00	R 30 000,00	R 30 000,00	R 30 000,00
Insurance	R 80 000,00	R 60 000,00	R 60 000,00	R 60 000,00	R 60 000,00	R 40 000,00
Salaries	R 600 000,00	R 600 000,00	R 600 000,00	R 600 000,00	R 600 000,00	R 600 000,00
Net income	R 328 914,40	R 2 182 337,31	R 7 653 332,50	R 19 096 354,12	R 26 241 137,76	R 26 792 713,07

Cash Flow Presentation

All prices are in „000

All revenues are rounded-up to the nearest „000



Learning Outcomes Evaluated

- Do you now have a clear understanding of what cost and revenue projection implies?
- Are you able to list and discuss some significant cost components commonly used in cost and revenue projections?
- Can you apply the “bottom-up” or “start to completion” approach, given a chain of operational activities, in cost and revenue projection?
- Can you now analyse and evaluate cost and revenue projections into the future, perhaps for 2, 3, 4, 5,.....10 to n years period?
- Can you now create, analyse, and evaluate best, most probable (optimal), and worst-case cost and revenue scenarios?
- Can you adapt this whole knowledge to the semester project?

Recommended reading in the handbook:

- Project Management - A Multi-disciplinary Approach (4th edition) Chapter 13: Pricing and Estimating
- Additional Alternative sources

The End

Next lecture

- Unit 11: Setting the MARR



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA